United States Patent Application Publication 20250287478
Kind Code A1
Publication Date September 11, 2025
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HEATING COOKER

Abstract

A heating cooker includes a heating and cooking chamber, a housing, a drawer body, a rack portion, and a pinion. A second opening is formed in a front wall of a housing. A drawer body can be pulled out along a second direction with respect to the heating and cooking chamber. A rack portion is inserted into a second opening and moves along the second direction together with the drawer body. A pinion rotates about an axis along a first direction and meshes with a rack tooth portion. The rack tooth portion is disposed at an end portion in a third direction of the rack portion. The rack portion includes a first wall portion disposed further on one side in the first direction than the rack tooth portion. An end portion on one side in the third direction of the first wall portion is positioned further on one side in the third direction than an end portion on one side in the third direction of the rack tooth portion.

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Family ID: **1000008419419**

Appl. No.: **19/021795**

Filed: **January 15, 2025**

Foreign Application Priority Data

JP	2024-036056	Mar. 08, 2024

Publication Classification

Int. Cl.: **H05B6/64** (20060101)

U.S. Cl.:

CPC **H05B6/6414** (20130101); **H05B6/6402** (20130101); H05B6/6485 (20130101)

Background/Summary

BACKGROUND OF THE INVENTION

1. Technical Field

[0001] The present disclosure relates to a heating cooker.

2. Description of the Related Art

[0002] JP 2008-8562 A discloses a drawer-type heating cooker. The drawer-type heating cooker disclosed in JP 2008-8562 A includes a magnetron that generates microwaves, a heating and cooking chamber, a drawer body that can be put in and taken out of the heating and cooking chamber, and a drive mechanism that drives the drawer body. The drive mechanism includes a pinion gear and a rack tooth portion (rack gear) meshing with the pinion gear.

SUMMARY OF THE INVENTION

[0003] In the drawer body of JP 2008-8562 A, since a tooth tip of the rack tooth portion protrudes, there is a possibility that the tooth tip of the rack tooth portion is damaged.

[0004] In view of the above problems, an object of the present disclosure is to provide a heating cooker capable of suppressing breakage of a tooth tip of a rack tooth portion.

[0005] According to one aspect of the present disclosure, a heating cooker includes a heating and cooking chamber, a housing, a drawer body, a rack portion, and a pinion. An object to be heated is accommodated in the heating and cooking chamber. The heating and cooking chamber is accommodated in the housing. The housing has a wall portion. An opening is formed in the wall portion. The drawer body can be drawn out along a first direction with respect to the heating and cooking chamber. The rack portion includes a rack tooth portion, extends in the first direction, is inserted into the opening, is connected to the drawer body, and is configured to move along the first direction together with the drawer body. The pinion is configured to rotate about an axis along a second direction intersecting the first direction and mesh with the rack tooth portion. The rack tooth portion is disposed at an end portion of the rack portion in a third direction intersecting the first direction and the second direction. The rack portion includes a first wall portion disposed further on one side or another side in the second direction than the rack tooth portion. An end portion on one side in the third direction of the first wall portion is positioned further on one side in the third direction than an end portion on one side in the third direction of the rack tooth portion.

[0006] According to the heating cooker of the present disclosure, it is possible to prevent a tooth tip of the rack tooth portions from being damaged.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1** is a perspective view of a heating cooker according to an embodiment of the present disclosure;

[0008] FIG. **2** is a perspective view of the heating cooker according to the present embodiment;

[0009] FIG. **3** is a view illustrating a left side surface of the heating cooker according to the present embodiment;

[0010] FIG. **4** is a perspective view illustrating the heating cooker in a state where a housing is removed according to the present embodiment;

[0011] FIG. **5** is a perspective view illustrating the heating cooker in a state where the housing is removed according to the present embodiment;

[0012] FIG. **6** is a view illustrating a left side surface of the heating cooker in a state where the housing is removed according to the present embodiment;

[0013] FIG. **7** is a perspective view illustrating a drive motor, a rack portion, and a pinion;

[0014] FIG. **8** is a perspective view illustrating a periphery of a second opening;
[0015] FIG. **9** is a front view illustrating a rack tooth portion and a pinion tooth portion; and
[0016] FIG. **10** is a view illustrating a variation of the rack portion.

DETAILED DESCRIPTION

[0017] Hereinafter, with reference to the drawings, an embodiment of a heating cooker according to the present disclosure will be described. Note that, in the drawings, the same or corresponding portions are denoted by the same reference numerals, and description of these will not be repeated.
[0018] With reference to FIGS. **1** and **2**, a heating cooker **100** according to the present embodiment will be described. FIGS. **1** and **2** are perspective views of the heating cooker according to an embodiment of the present disclosure. To be specific, FIG. **1** illustrates the external appearance of the heating cooker **100** when viewed diagonally from the lower left front. FIG. **2** illustrates the external appearance of the heating cooker **100** when viewed diagonally from the upper right rear. As illustrated in FIGS. **1** and **2**, the heating cooker **100** is a drawer-type heating cooker. The heating cooker **100** heats and cooks an object to be heated. The object to be heated is, for example, a food item. The heating cooker **100** includes a housing **10**, a drawer body **20**, and an operation panel **30**.
[0019] The operation panel **30** is a substantially rectangular plate-shaped member. The operation panel **30** receives operation from the user. The operation includes, for example, a cooking method for heating and cooking an object to be heated. Specifically, the operation panel **30** includes a display unit and a plurality of buttons. The display unit displays various types of information. Specifically, the display unit includes a liquid crystal panel.
[0020] In the present embodiment, a side of the heating cooker **100** on which the operation panel **30** is disposed is defined as a front side of the heating cooker **100**, and a side (back surface side) opposite to the front side is defined as a rear side of the heating cooker **100**. Further, when the heating cooker **100** is viewed from the front side, a right side is defined as a right side of the heating cooker **100**, and a side opposite to the right side is defined as a left side of the heating cooker **100**. Further, in a direction orthogonal to a front-rear direction and a left-right direction of the heating cooker **100**, a side on which the operation panel **30** is disposed is defined as an upper side of the heating cooker **100**, and a side (bottom side) opposite to the upper side is defined as a lower side of the heating cooker **100**. Note that, these directions are not intended to limit directions when the heating cooker **100** of the present disclosure is used. In the present embodiment, a first direction D**1** is an upward direction. A second direction D**2** is a forward direction. A third direction D**3** is a left direction.
[0021] The housing **10** is a box-shaped member. Specifically, the housing **10** has a right outer wall **11**, a left outer wall **12**, an upper outer wall **13**, a lower outer wall **14**, a rear outer wall **15**, a front wall **16**. The rear outer wall **15** intersects the second direction D**2**. The right outer wall **11** and the left outer wall **12** face each other in the third direction D**3**. The upper outer wall **13** and the lower outer wall **14** face each other in the first direction D**1**. The housing **10** accommodates a heating and cooking chamber **50** to be described below.
[0022] Next, the heating and cooking chamber **50** will be described with reference to FIGS. **3** to **6**. FIG. **3** is a view illustrating a left side surface of the heating cooker **100** according to the present embodiment. FIGS. **4** and **5** are perspective views illustrating the heating cooker **100** in a state where the housing **10** is removed according to the present embodiment. To be specific, FIG. **4** illustrates the external appearance of the heating cooker **100** when viewed diagonally from the lower left front. FIG. **5** illustrates the external appearance of the heating cooker **100** when viewed diagonally from the upper right rear. FIG. **6** is a view illustrating a left side surface of the heating cooker **100** in a state where the housing **10** is removed according to the present embodiment.
[0023] As illustrated in FIGS. **1** and **3**, the heating and cooking chamber **50** is accommodated in the housing **10**. The heating and cooking chamber **50** accommodates an object to be heated. The heating and cooking chamber **50** has, for example, a substantially rectangular parallelepiped shape. Specifically, as illustrated in FIGS. **4** to **6**, the heating and cooking chamber **50** has a right wall **51**,

a left wall **52**, an upper wall **53**, a lower wall **54**, and a rear wall **55**. The rear wall **55** intersects the second direction D**2**. The right wall **51** and the left wall **52** face each other in the third direction D**3**. The upper wall **53** and the lower wall **54** face each other in the first direction D**1**. A material of each of the right wall **51**, the left wall **52**, the upper wall **53**, the lower wall **54**, and the rear wall **55** is, for example, stainless steel. Note that the material of each wall is not limited to stainless steel. Any metal having at least predetermined heat resistance and reflectance characteristics may be used.

[0024] A front end of the right wall **51**, the left wall **52**, the upper wall **53**, and the lower wall **54** is connected to a rear surface of the front wall **16**. A rear end of the right wall **51**, the left wall **52**, and the lower wall **54** is connected to a front surface of the rear wall **55**. An upper end portion of the rear wall **55** is connected to a lower surface of the upper wall **53**, and a lower end portion of the rear wall **55** is connected to an upper surface of the lower wall **54**.

[0025] As illustrated in FIG. **4**, the front wall **16** includes a first opening **161**, a second opening **162**, a third opening **163**, a suction port **164**, and a blow-out port **165**. In other words, the housing **10** has the front wall **16** in which the first opening **161** and the second opening **162** are formed. The front wall **16** corresponds to, for example, a “wall portion”. Further, the first opening **161** corresponds to, for example, “another opening”, and the second opening **162** corresponds to, for example, an “opening”.

[0026] An object to be heated passes through the first opening **161**. The first opening **161** is formed in a substantially rectangular shape. The first opening **161** is positioned inside a region defined by a connection portion between the right wall **51**, the left wall **52**, the upper wall **53**, and the lower wall **54**, and the front wall **16**.

[0027] The second opening **162** is positioned at an intermediate position and below the first opening **161** in the left-right direction of the first opening **161**. Specifically, the second opening **162** is positioned between the lower outer wall **14** and the lower wall **54**. The second opening **162** is formed in a substantially rectangular shape. The second opening **162** includes a notch-shaped opening having no partition portion with respect to the first opening **161**.

[0028] For example, a pair of left and right ones of the third openings **163** are provided. A pair of the third openings **163** are disposed at a position corresponding to an intermediate position of the first opening **161** in an up-down direction. Specifically, the third opening **163** is disposed between the upper outer wall **13** and the lower outer wall **14**. Further, a pair of the third openings **163** are positioned with the first opening **161** interposed between them in the third direction D**3**. Note that the third opening **163** includes a notch-shaped opening having no partition portion with respect to the first opening **161**.

[0029] Air passes through the suction port **164** from the outside of the housing **10** toward the inside of the housing **10**. For example, a plurality of the suction ports **164** are provided. A plurality of the suction ports **164** are disposed at a lower position of the first opening **161** and a lower position of the second opening **162**.

[0030] Air passes through the blow-out port **165** from the inside of the housing **10** toward the outside of the housing **10**. For example, a plurality of the blow-out ports **165** are provided. A plurality of the blow-out ports **165** is disposed at an upper position than the first opening **161**.

[0031] As illustrated in FIGS. **4** and **5**, the rear wall **55** has a fourth opening **551**. Air flowing in through the suction port **164** passes through the fourth opening **551** to the rear side of the rear wall **55**. For example, a plurality of the fourth openings **551** are provided. The fourth opening **551** is disposed at a corresponding position between the lower outer wall **14** and the lower wall **54**.

[0032] As illustrated in FIGS. **4** to **6**, the heating cooker **100** further includes a first space R**1**, a second space R**2**, a third space R**3**, a fourth space R**4**, and a fifth space R**5**. The second space R**2** is disposed between the lower outer wall **14** and the lower wall **54**. The fourth space R**4** is disposed between the right outer wall **11** and the right wall **51**. The fifth space R**5** is disposed between the left outer wall **12** and the left wall **52**.

[0033] As illustrated in FIGS. **5** and **6**, the first space R**1** is positioned between the upper outer wall **13** and the upper wall **53**. In the first space R**1**, a motor **31** and a waveguide **32** are disposed. The motor **31** rotates an antenna of a magnetron **33** that is a microwave generation device. The waveguide **32** propagates microwaves generated by the magnetron **33**. The heating cooker **100** executes a microwave heating mode by operation of the magnetron **33**.
[0034] The third space R**3** is disposed between the rear outer wall **15** and the rear wall **55**. The magnetron **33**, a control board **34**, a high-voltage transformer **35**, a high-voltage capacitor **36**, and a cooling fan **37** are disposed in the third space R**3**. The control board **34**, the high-voltage transformer **35**, and the high-voltage capacitor **36** are fixed to an upper portion of the lower outer wall **14** along the third direction D**3**. The cooling fan **37** is disposed above the high-voltage transformer **35** adjacent to the left outer wall **12**. The cooling fan **37** sucks air from the suction port **164** into the second space R**2**. Air sucked into the second space R**2** flows into the third space R**3** through the fourth opening **551**. Air flowing into the third space R**3** cools electric components such as the high-voltage transformer **35**, and a part of the air passes through the heating and cooking chamber **50**, is sent to the first space R**1**, and is blown out of the housing **10** through the blow-out port **165**.
[0035] Next, the drawer body **20** will be described with reference to FIGS. **1** to **6**. The drawer body **20** can be pulled out from the heating and cooking chamber **50**. The drawer body **20** opens the first opening **161** by movement in the second direction D**2**, and closes the first opening **161** by movement in a direction opposite to the second direction D**2**. As illustrated in FIGS. **1** to **6**, the drawer body **20** includes a door **21**, left and right slide portions **22**, a lower slide portion **61**, and a placement portion **23**. The second direction D**2** corresponds to, for example, “one side in the first direction”. A direction opposite to the second direction D**2** corresponds to, for example, “another side in the first direction”.
[0036] The door **21** is configured to be able to open and close the first opening **161**. The suction port **164** and the blow-out port **165** are open also in a state where the door **21** closes the first opening **161**.
[0037] As illustrated in FIGS. **4** to **6**, the slide portion **22** includes a movable rail **22***a* and a fixed rail **22***b*. The slide portion **22** is a telescopic slide mechanism in which the movable rail **22***a* is slidable with respect to the fixed rail **22***b*.
[0038] For example, a pair of the fixed rails **22***b* are provided on the left and right. A pair of the fixed rails **22***b* are disposed in each of the fourth space R**4** and the fifth space R**5**. A pair of the fixed rails **22***b* are fixed to an outer surface of the right wall **51** and an outer surface of the left wall **52**. Specifically, the fixed rail **22***b* is disposed along the second direction D**2** at an intermediate position in the up-down direction of the right wall **51** (left wall **52**).
[0039] For example, a pair of the movable rails **22***a* are provided on the left and right. Front end portions of a pair of the movable rails **22***a* are connected to a rear surface of left and right end portions of the door **21**. A pair of the movable rails **22***a* extend in the second direction D**2** and is inserted into a pair of the third openings **163**. When the drawer body **20** is completely pulled out from the heating and cooking chamber **50**, length of the movable rail **22***a* extending from the third opening **163** in the second direction D**2** is longest. The first opening **161** is opened. On the other hand, when the drawer body **20** is completely pulled into the heating and cooking chamber **50**, length of the movable rail **22***a* extending from third opening **163** in second direction D**2** is shortest. The first opening **161** is closed by the door **21**.
[0040] An object to be heated is placed on the placement portion **23**. The placement portion **23** includes a rear wall portion **23***b*, left and right side wall portions **23***c*, and a bottom wall portion **23***d*. Front end portions of the left and right side wall portions **23***c* are connected to a rear surface of the door **21**, and rear end portions of the left and right side wall portions **23***c* are connected to a front surface of the rear wall portion **23***b*. Lower end portions of the left and right side wall portions **23***c* and the rear wall portion **23***b* are connected to the bottom wall portion **23***d*. Further,

the placement portion **23** is connected to the movable rail **22***a* via the door **21**. Therefore, when the movable rail **22***a* is pulled out from the third opening **163**, the placement portion **23** is pulled out from the heating and cooking chamber **50** in the second direction D**2** along with movement of the movable rail **22***a*.

[0041] The heating cooker **100** further includes a moving mechanism **200**. The moving mechanism **200** moves the drawer body **20** forward and backward along the second direction D**2**.

[0042] The moving mechanism **200** will be described with reference to FIGS. **7** to **9**. FIG. **7** is a perspective view illustrating a drive motor **40**, a rack portion **60**, and a pinion **70**. FIG. **8** is a perspective view illustrating a periphery of the second opening **162**. FIG. **9** is a front view illustrating a rack tooth portion **62** and a pinion tooth portion **72**. The moving mechanism **200** includes the drive motor **40**, the rack portion **60**, and the pinion **70**.

[0043] The rack portion **60** moves along the front-rear direction together with the drawer body **20**. As illustrated in FIGS. **1**, **4**, and **8**, the lower slide portion **61** having the rack portion **60** extends along the front-rear direction and is inserted into the second opening **162**. A front end portion of the slide portion **61** is connected to a rear surface of a lower portion of the door **21** at a center portion in the left-right direction. In other words, the slide portion **61** having the rack portion **60** is connected to the drawer body **20**. As illustrated in FIGS. **7** to **9**, the rack portion **60** includes the rack tooth portion **62** and a first wall portion **63**.

[0044] The rack tooth portion **62** is disposed at an end portion of the rack portion **60** in the left-right direction. A plurality of the rack tooth portions **62** are provided at an end portion **61***a* of the slide portion **61** described later in the third direction D**3**. The rack tooth portion **62** extends in the third direction D**3** from the end portion **61***a* of the slide portion **61** in the third direction D**3**. A plurality of the rack tooth portions **62** are aligned along the second direction D**2**.

[0045] The pinion **70** meshes with the rack tooth portion **62**. The pinion **70** rotates about an axis **74** along the up-down direction.

[0046] The first wall portion **63** is disposed at an end portion of the rack portion **60** in the third direction D**3**. The first wall portion **63** is disposed on one side or another side of the rack tooth portion **62** in the up-down direction. Specifically, the first wall portion **63** is disposed at a position opposite to the rack tooth portion **62** in the first direction D**1**. Note that the first direction D**1** corresponds to, for example, “one or another side of the second direction”.

[0047] An end portion **63***a* on one side in the left-right direction of the first wall portion **63** is positioned further on one side in the left-right direction than an end portion **62***a* on one side in the left-right direction of the rack tooth portion **62**. Specifically, the end portion **63***a* in the third direction D**3** of the first wall portion **63** is positioned further in the third direction D**3** than the end portion **62***a* in the third direction D**3** of the rack tooth portion **62**. Since the end portion **63***a* in the third direction D**3** of the first wall portion **63** is positioned further in the third direction D**3** than the end portion **62***a* in the third direction D**3** of the rack tooth portion **62**, it is possible to prevent the end portion **62***a* in the third direction D**3** of the rack tooth portion **62** from protruding in the third direction D**3** from the end portion **63***a* in the third direction D**3** of the first wall portion **63**. By the above, the rack tooth portion **62** can be prevented from being damaged.

[0048] The lower slide portion **61** includes the rack portion **60**, a movable rail, and a fixed rail. The slide portion **61**, the movable rail, and the fixed rail are made from metal. The rack portion **60** includes the rack tooth portion **62**, the first wall portion **63**, and a second wall portion **64**. The rack portion **60** is made from synthetic resin and may be integrally formed. In the present embodiment, slide portions are disposed on both the left and right sides and on the lower side, and the rack portion **60** is disposed on the lower slide portion **61**, but the configuration is not limited to this. The rack portion **60** may be provided on at least one of the left and right slide portions **22**. Note that, in consideration of balanced forward and backward movement, it is preferable to arrange the rack portion **60** on the lower slide portion **61**.

[0049] As illustrated in FIGS. **7** to **9**, the lower slide portion **61** extends along the second direction

D**2**. The slide portion **61** is formed from a plate-like member. The slide portion **61** moves along a slide rail **66** (see FIG. **6**). As illustrated in FIG. **6**, the slide rail **66** is fixed in a state of being stretched between the front wall **16** and the rear wall **55** along the second direction D**2**.

[0050] As illustrated in FIGS. **7** to **9**, the rack tooth portion **62**, the first wall portion **63**, and the second wall portion **64** are provided at an end portion of the slide portion **61** in the third direction D**3**. The slide portion **61** constitutes a groove portion **65** together with the first wall portion **63** and the second wall portion **64**.

[0051] The second wall portion **64** is disposed opposite to the first wall portion **63** with respect to the rack tooth portion **62**. Specifically, the second wall portion **64** is disposed with the rack tooth portion **62** interposed between the first wall portion **63** and the second wall portion **64**. An end portion **64***a* in the third direction D**3** of the second wall portion **64** is positioned further in the third direction than the end portion **62***a* in the third direction D**3** of the rack tooth portion **62**. In the present embodiment, the end portion **63***a* of the first wall portion **63** and the end portion **64***a* of the second wall portion **64** are positioned at the same position in the third direction D**3**. As described above, the rack tooth portion **62** can be accommodated in the groove portion **65** formed by the first wall portion **63** and the second wall portion **64**. By the above, the end portion **62***a* in the third direction D**3** of the rack tooth portion **62** can be prevented from protruding in the third direction D**3** from the end portion **63***a* in the third direction D**3** of the first wall portion **63**, and the end portion **62***a* in the third direction D**3** of the rack tooth portion **62** can be prevented from protruding in the third direction D**3** from the end portion **64***a* in the third direction D**3** of the second wall portion **64**. Accordingly, it is possible to further prevent the rack tooth portion **62** from being damaged.

[0052] As illustrated in FIGS. **4** and **6**, the rack portion **60** is disposed below the placement portion **23**. Since the rack portion **60** is disposed below the placement portion **23**, it is possible to prevent approach from the placement portion **23** side to the rack tooth portion **62** by being blocked by the placement portion **23**. As a result, damage to the rack tooth portion **62** can be suppressed. Since the rack portion **60** is disposed below the placement portion **23**, the rack portion **60** is blocked by the placement portion **23** and is not visible to the user. As a result, deterioration in appearance can be suppressed.

[0053] As illustrated in FIGS. **7** to **9**, the first wall portion **63** is disposed below the rack tooth portion **62**. The placement portion **23** can prevent approach toward the rack tooth portion **62** from above, and the first wall portion **63** can prevent approach toward the rack tooth portion **62** from below.

[0054] The rack tooth portion **62** is disposed at the end portion **61***a* of the slide portion **61** in a horizontal direction. Since an upper surface of the rack portion **60** can be disposed to face the bottom wall portion **23***d* of the placement portion **23**, approach to the rack tooth portion **62** from above can be further prevented. Further, since the rack portion **60** and the bottom wall portion **23***d* are disposed close to each other, the heating cooker **100** can be downsized. The rack portion **60** is driven by the drive motor **40**.

[0055] As illustrated in FIG. **7**, the drive motor **40** includes a first pulley **41** and a clutch unit **42**. The drive motor **40** is disposed in the second space R**2**, for example, and is fixed to a rear surface of the front wall **16**. For example, in a case where an opening operation button (not illustrated) of the operation panel **30** is turned on, the drive motor **40** moves the drawer body **20** by forward rotation to open the first opening **161**. For example, in a case where a closing operation button (not illustrated) of the operation panel **30** is turned on, the drive motor **40** moves the drawer body **20** by reverse rotation to close the first opening **161**.

[0056] The first pulley **41** is fixed to a tip of a drive shaft of the drive motor **40**. Specifically, the first pulley **41** is fixed to a lower end of the drive shaft of the drive motor **40**. The first pulley **41** transmits driving force to the pinion **70** via a belt **43**.

[0057] The clutch unit **42** transmits or cuts off driving force from the drive motor **40** toward the first pulley **41**. The clutch unit **42** is disposed between a motor main body and the first pulley **41**.

For example, when an opening operation button of the operation panel **30** is turned on, since the clutch unit **42** is connected, driving force is transmitted from the drive motor **40** to the first pulley **41**. The clutch unit **42** is cut off at a timing when pull-out of the drawer body **20** is completed, and driving force from the drive motor **40** toward the first pulley **41** is cut off. For example, when a close operation button of the operation panel **30** is turned on, the clutch unit **42** is connected, so that driving force is transmitted from the drive motor **40** to the first pulley **41**. In the clutch unit **42**, the clutch unit **42** is cut off at a timing when pull-in of the drawer body **20** is completed, and driving force from the drive motor **40** toward the first pulley **41** is cut off. Here, the drive motor **40** does not need to include the clutch unit **42**. In a case where the drive motor **40** does not include the clutch unit **42**, for example, when an opening operation button of the operation panel **30** is turned on, a drive shaft of the drive motor **40** is operated, and the drive shaft of the drive motor **40** only needs to be stopped at a timing when pull-out of the drawer body **20** is completed. For example, the same applies when a closing operation button of the operation panel **30** is turned on.

[0058] The pinion **70** transmits driving force input to a second pulley **73** via the belt **43** to the rack portion **60**. A spline formed on an inner periphery of the pinion **70** is engaged with a spline formed on a rotation shaft of the second pulley **73**. Specifically, the pinion **70** is connected to an upper portion of a rotation shaft extending upward from the second pulley **73**.

[0059] The pinion **70** is made from metal. Since the rack portion **60** is made from synthetic resin, the rack portion **60** can elastically deform to follow the pinion **70**. By the above, it is possible to suppress squeaking noise generated at the time of meshing between the pinion **70** and the rack portion **60** while suppressing tooth skipping of the pinion **70**. However, the pinion **70** may be made from synthetic resin.

[0060] As illustrated in FIGS. **7** and **9**, the pinion **70** includes a shaft portion **71** and the pinion tooth portion **72**.

[0061] The pinion tooth portion **72** extends radially outward from the shaft portion **71** and meshes with the rack tooth portion **62**. An upper end portion **721** of the pinion tooth portion **72** is positioned below an upper end portion **621** of the rack tooth portion **62**. A lower end portion **722** of the pinion tooth portion **72** is positioned above a lower end portion **622** of the rack tooth portion **62**. By the above, during opening operation of the drawer body **20**, it is possible to suppress interference between the pinion tooth portion **72** and the first wall portion **63** or the second wall portion **64**, and it is possible to suppress generation of sound caused by the interference.

[0062] An upper end portion **711** of the shaft portion **71** is positioned above an upper end portion **621** of the rack tooth portion **62**. A lower end portion **712** of the shaft portion **71** is positioned below a lower end portion **622** of the rack tooth portion **62**. By the above, rigidity of the shaft portion **71** can be increased by enlarging the shaft portion **71**, so that a posture of the pinion **70** can be maintained and occurrence of tooth skipping can be suppressed.

[0063] The embodiment of the present disclosure is described above with reference to the drawings (FIGS. **1** to **9**). However, the present disclosure is not limited to the above embodiment, and can be implemented in various aspects without departing from the gist of the present disclosure (for example, (1) to (7) described below). For easy understanding, the drawings schematically illustrate each constituent element mainly, and thickness, length, the number, and the like of constituent elements illustrated in the drawings are different from actual ones for convenience of preparation of the drawings. Further, materials, shapes, dimensions, and the like of constituent elements illustrated in the above embodiment are merely examples, and are not particularly limited, and various modifications can be made without substantially departing from the effects of the present disclosure. [0064] (1) As described with reference to FIGS. **1** to **9**, the rack tooth portion **62** is disposed on one side in the horizontal direction, and a surface portion of the rack portion **60** intersects a vertical direction. However, the present disclosure is not limited to this. The rack portion **60** may be disposed such that the rack tooth portion **62** is arranged on one side in the vertical direction and a surface portion of the rack portion **60** intersects the horizontal direction.

[0065] (2) As described with reference to FIGS. **1** to **9**, the rack portion **60** is disposed such that a surface portion of the rack portion **60** faces the bottom wall portion **23***d* of the placement portion **23**. However, the present disclosure is not limited to this. The rack portion **60** may be disposed such that a surface portion of the rack portion **60** faces the side wall portion **23***c* of the placement portion **23**. [0066] (3) As described with reference to FIGS. **1** to **9**, the drive motor **40** is fixed to a rear surface of the front wall **16**, and the first pulley **41** is fixed to a lower end of a drive shaft of the drive motor **40**, but the present disclosure is not limited to this. The drive motor **40** may be fixed to an upper surface of the lower outer wall **14**, and the first pulley **41** may be fixed to an upper end of a drive shaft of the drive motor **40**. With this configuration, the second pulley **73** is rotated by the first pulley **41** fixed to an upper end of a drive shaft of the drive motor **40**, and the pinion **70** connected to an upper portion of a rotation shaft extending upward from the second pulley **73** is rotated. [0067] (4) As described with reference to FIGS. **1** to **9**, the end portion **63***a* of the first wall portion **63** and the end portion **64***a* of the second wall portion **64** are positioned at the same position in the third direction D**3**, but the present disclosure is not limited to this. The end portion **63***a* of the first wall portion **63** and the end portion **64***a* of the second wall portion **64** may be positioned at different positions in the third direction D**3**.

[0068] As illustrated in FIG. **10**, the end portion **63***a* in the third direction D**3** of the first wall portion **63** disposed lower between the first wall portion **63** and the second wall portion **64** may be positioned further in the third direction D**3** than the end portion **64***a* in the third direction D**3** of the second wall portion **64**. With this configuration, since the end portion **63***a* in the third direction D**3** of the first wall portion **63** is positioned further in the third direction D**3** than the end portion **64***a* in the third direction D**3** of the second wall portion **64**, the second wall portion **64** can be made small while catching of a finger and the like is suppressed. As a result, the rack portion **60** can be downsized. [0069] (5) As described with reference to FIGS. **1** to **9**, the rack tooth portion **62**, the first wall portion **63**, and the second wall portion **64** are formed as a single member, but the present disclosure is not limited to this. The rack tooth portion **62**, the first wall portion **63**, and the second wall portion **64** may be formed as separate members, and these members may be connected via a connection member. Further, at least any of the rack tooth portion **62**, the first wall portion **63**, and the second wall portion **64** may be made from metal. For example, the rack tooth portion **62** and the second wall portion **64** may be formed as a single member, an end portion on one side in the third direction D**3** of the slide portion **61** may be positioned further on one side in the third direction D**3** than an end portion on one side in the third direction D**3** of the rack tooth portion **62**, and at least a part of the slide portion **61** may be the first wall portion **63**. Further, the rack tooth portion **62** and the first wall portion **63** may be formed as a single member, and an end portion on one side in the third direction D**3** of the slide portion **61** may be positioned further on one side in the third direction D**3** than an end portion on one side in the third direction D**3** of the rack tooth portion **62**, so that at least a part of the slide portion **61** is configured to be the second wall portion **64**. [0070] (6) As described with reference to FIGS. **1** to **9**, the lower slide portion **61** is configured to include the rack portion **60**, but the present disclosure is not limited to this. At least one of the left and right slide portions **22** may be configured to have a rack portion. The moving mechanism **200** only needs to be configured to support the rack portion **60**. [0071] (7) As described with reference to FIGS. **1** to **9**, the heating cooker **100** performs only a microwave heating mode as a heating cooking mode, but the present disclosure is not limited to this. The heating cooker **100** may execute other cooking modes such as a grill heating mode and a hot air circulation heating mode in addition to the microwave heating mode as the heating cooking mode. The grill heating mode is a mode for heating and cooking an object to be heated mainly by exposing the object to be heated to radiant heat. The hot air circulation heating mode is a mode for heating and cooking an object to be heated mainly by circulating hot air in the inside of the heating and cooking chamber **50** to achieve uniform temperature in the heating and cooking chamber **50**.

[0072] The present disclosure is useful, for example, in the field of heating cookers.

Claims

1. A heating cooker comprising: a heating and cooking chamber in which an object to be heated is accommodated; a housing in which the heating and cooking chamber is accommodated, the housing having a wall portion in which an opening is formed; a drawer body that can be pulled out along a first direction with respect to the heating and cooking chamber; a rack portion that has a rack tooth portion, extends in the first direction, is inserted into the opening, is connected to the drawer body, and is configured to move along the first direction together with the drawer body; and a pinion configured to rotate about an axis along a second direction intersecting the first direction and mesh with the rack tooth portion, wherein the rack tooth portion is disposed at an end portion in a third direction intersecting the first direction and the second direction of the rack portion, the rack portion includes a first wall portion disposed further on one side or another side in the second direction than the rack tooth portion, and an end portion on one side in the third direction of the first wall portion is positioned further on one side in the third direction than an end portion on one side in the third direction of the rack tooth portion.
2. The heating cooker according to claim 1, wherein the rack portion includes a second wall portion disposed at a position opposite to the first wall portion with respect to the rack tooth portion, and an end portion on one side in the third direction of the second wall portion is positioned further on one side in the third direction than an end portion on one side in the third direction of the rack tooth portion.
3. The heating cooker according to claim 2, wherein the pinion includes: a shaft portion; and a pinion tooth portion that extends radially outward from the shaft portion and is configured to mesh with the rack tooth portion, an end portion on one side in the second direction of the shaft portion is positioned further on one side in the second direction than an end portion on one side in the second direction of the rack tooth portion, an end portion on another side in the second direction of the shaft portion is positioned further on another side in the second direction than an end portion on another side in the second direction of the rack tooth portion, an end portion on one side in the second direction of the pinion tooth portion is positioned further on another side in the second direction than an end portion on one side in the second direction of the rack tooth portion, and an end portion on another side in the second direction of the pinion tooth portion is positioned further on one side in the second direction than an end portion on another side in the second direction of the rack tooth portion.
4. The heating cooker according to claim 1, wherein the wall portion further includes another opening different from the opening, the drawer body includes: a door configured to be capable of opening and closing the another opening; and a placement portion on which the object to be heated is placed, and the rack portion is disposed below the placement portion.
5. The heating cooker according to claim 2, wherein the drawer body includes: a placement portion on which the object to be heated is placed; and a slide portion disposed below the placement portion, and the rack tooth portion is disposed at an end portion in a horizontal direction of the slide portion.
6. The heating cooker according to claim 4, wherein the second direction indicates a direction along a vertical direction, and the first wall portion is disposed below the rack tooth portion.
7. The heating cooker according to claim 4, wherein the rack portion includes a second wall portion disposed opposite to the first wall portion with respect to the rack tooth portion, an end portion on one side in the third direction of the second wall portion is positioned further on one side in the third direction than an end portion on one side in the third direction of the rack tooth portion, and an end portion on one side in the third direction of one wall portion disposed lower between the first wall portion and the second wall portion is positioned further on one side in the third direction than an end portion on one side in the third direction of another wall portion.

8. The heating cooker according to claim 1, wherein the pinion is made from metal, and the rack portion is made from synthetic resin.